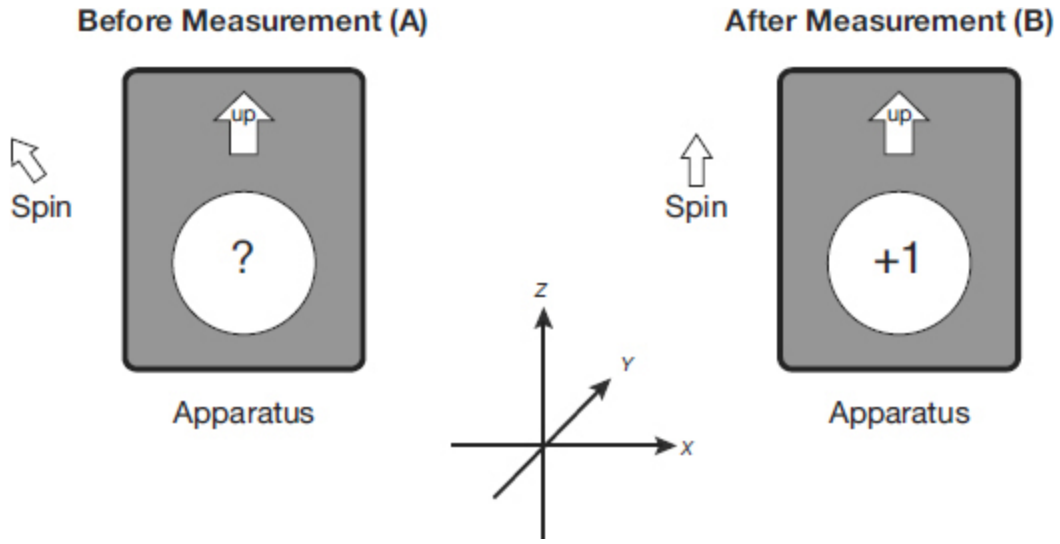


Figure 1.1: (A) Spin and cat-free apparatus before any measurement is made. (B) Spin and apparatus after one measurement has been made, resulting in $\sigma_z = +1$. The spin is now prepared in the $\sigma_z = +1$ state. If the spin is not disturbed and the apparatus keeps the same orientation, all subsequent measurements will give the same result. Coordinate axes show our convention for labeling the directions of space.

Now that we've measured σ, let's reset the apparatus to neutral and, without disturbing the spin, measure σ again. Assuming the simple law of Eq. 1.1, we should get the same answer as we did the first time. The result $\sigma = +1$ will be followed by $\sigma = +1$. Likewise for $\sigma = -1$. The same will be true for any number of repetitions. This is good because it allows us to confirm the result of an experiment. We can also say this in the following way: The first interaction with the apparatus $\mathcal{A}$ *prepares* the system in one of the two states. Subsequent experiments *confirm* that state. So far, there is no difference between classical and quantum physics.